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Oefeningenreeks 2D nr. 12.15

$$
\begin{aligned}
&= \lim_{x\to\infty} \left(\sqrt[3]{x^3+6x^2} - x\right) \\
&= \lim_{x\to\infty} \frac{\left(\sqrt[3]{x^3+6x^2} - x\right)\left(\sqrt[3]{x^3+6x^2}^{\,2} + x \cdot \sqrt[3]{x^3+6x^2} + x^2\right)}{\sqrt[3]{x^3+6x^2}^{\,2} + x \cdot \sqrt[3]{x^3+6x^2} + x^2} \\
&= \lim_{x\to\infty} \frac{x^3+6x^2-x^3}{\sqrt[3]{x^3+6x^2}^{\,2} + x \cdot \sqrt[3]{x^3+6x^2} + x^2} \\
&= \lim_{x\to\infty} \frac{6x^2}{\sqrt[3]{x^3+6x^2}^{\,2} + x \cdot \sqrt[3]{x^3+6x^2} + x^2} \\
&= \frac{6}{3} = 2
\end{aligned}
$$

References